



Fuel Not Found Hackathon 2026

Hackathon Participant Handbook

EV Dashboard: Smart Carbon Savings and Eco-Driving Assistant

1. Introduction

As electric vehicles (EVs) become increasingly popular, many users purchase EVs with the intention of contributing towards environmental sustainability. However, most EV drivers have limited visibility into the actual environmental impact of their daily driving activities.

Current EV dashboards primarily focus on operational information such as battery percentage, vehicle speed, remaining range, and charging status. While these indicators are important for vehicle operation, they do not effectively communicate the environmental benefits achieved through EV usage.

The proposed EV Dashboard aims to address this gap by providing users with meaningful sustainability insights, including real-time carbon emissions savings, comparisons with conventional petrol vehicles, and recommendations for eco-friendly driving behaviour. The project integrates software development, embedded systems, data analytics, and sustainability awareness into a single platform that aligns with the objectives of the Fuel Not Found Hackathon.

2. Problem Statement

The hackathon challenge highlights that EV users often lack awareness of the actual carbon savings achieved through their vehicle usage.

Several issues have been identified:

1. EV users cannot easily visualize their contribution to carbon reduction.
2. Existing dashboards focus mainly on vehicle performance rather than sustainability metrics.
3. Drivers may unknowingly adopt inefficient driving habits, leading to increased energy consumption.
4. Limited feedback is available to encourage eco-friendly driving behaviour.

As a result, the environmental advantages of EV adoption are not fully communicated to users, reducing opportunities to promote sustainable driving practices.

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3. Expected Benefits

Environmental

- Promote sustainability awareness among EV users.
- Encourage environmentally responsible driving behaviour.

Educational

- Improve understanding of carbon emissions and energy consumption.

Technical

- Demonstrate integration between hardware and software systems.
- Showcase IoT and data analytics applications within EV technology.

4. Workshop (Pre-Hackathon Training Session)

A technical workshop will be conducted prior to the hackathon to support participants in preparing for the hardware component. The workshop will provide foundational knowledge in embedded systems, focusing on Arduino development.

Workshop speaker: Ts Dr Lee Han Kee

Date: 29/6/2026, Monday

Time: 2:00 PM – 3:00 PM

Venue: FEGT (exact venue will be announce soon)

Hands on: Arduino-based hands-on session

5. Hackathon rules and regulations

Eligibility	Student Status - Open exclusively to currently enrolled Universiti Tunku Abdul Rahman (UTAR) students - Only students from FICT (UTAR KAMPAR), FEGT (UTAR KAMPAR), LKCFES (UTAR SG LONG) faculties are eligible - All academic years are eligible - Participants may be required to present a valid student ID Teams - Teams must consist of 3-5 members - Each students may only join one team - No team changes after hackathon commencement - No funding will be provided for project development Judging - All judges' decisions are final - No appeals will be entertained Code ownership / originality rule - All submissions must be original work created during the hackathon period - Plagiarism, pre-built projects, or external unfair assistance will result in disqualification Disqualification rules: • Plagiarism • Pre-built projects • External unfair assistance • Rule violation - Violation of rules may result in immediate disqualification
Project Rules & Academic Integrity	Project Integrity - Projects must be relevant to the hackathon theme - All coding, hardware assembly, and design must be completed during the hackathon period - Ideas may be discussed beforehand; however, all coding, hardware assembly, and design work must be completed during the hackathon period - Reuse of past coursework, capstone projects, or existing personal projects is strictly prohibited Use of Tools & Learning Resources - Open-source libraries and frameworks, public APIs, AI tools, online tutorials and documentation are allowed given that - All teams members must understand and explain all parts of their project - Direct plagiarism or blind copying is strictly prohibited - AI tools are allowed for learning and reference purposes only. Teams must be able to fully explain and justify their implementation
Projects	Proposal (Preliminary round) - No citation needed - Proposal limited to 5 pages only excluding cover page which contains project name, team name - Report format as follows: Font: Times New Roman Font Size: 12 pt

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Projects	Line Spacing: 1.5 Margins: Normal (1 inch / 2.54 cm all around) Alignment: Justified Page Numbers: Bottom center - Submission 29 June 2026, Monday by 12:00 AM - Any late submission will be considered as a withdrawal from the hackathon. - Submission file format supported is PDF - File naming: GroupName_Proposal - Submission email to: mbotfuelnotfound@gmail.com Software projects (Final round) - Must include a live working demo during presentation - Core functionality implemented during the hackathon period Hardware projects (Final round) - Must use the provided ESP32 microcontroller - Must present a functional prototype or proof-of-concept - Must clearly explain working and simulated components - External tools, materials, or APIs will not be funded by organisers
Time & Development Rules	Time - Development begins only after official hackathon commencement and after finalist announcement - All coding, designing, and building must stop at submission deadline - Late submissions may be disqualified
Submission Requirements	Proposal submission requirement: 1) project name & team name 2) problem statement (may be refined from this handbook) 3) solution description 4) expected demo: live demonstration 5) technical explanation with diagram 6) impact and targeted users 7) future improvements 8) conclusion *Proposal topics are not limited to the example provided in this handbook. Teams are encouraged to propose their own ideas, as long as they are relevant to the hackathon theme.
Presentation	- Finalist teams are required to present their project during the final round Each team will be given 15–20 minutes in total, including presentation and live demo Presentation must clearly cover: • Problem statement • Proposed solution • System features • Technical implementation • Sustainability impact • Future improvements - A live demonstration is required, while recorded demo may be used as backup - All team members must participate and demonstrate understanding of the project - A Q&A session will be conducted after each presentation, subject to judges' discretion - Judges' decisions are final

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6. Hackathon winning criteria (Final Round)

Problem Definition & Understanding (15 marks)
Creativity & Innovation (20 marks)
Technical Execution (25 marks)
Presentation & Communication (20 marks)
Sustainability & Impact (20 marks)

7. Breakdown of Report

Project name Team name	Page 1	Page 2
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8. Conclusion

The EV Dashboard proposes a practical solution to improving sustainability awareness among EV users by translating vehicle data into meaningful environmental insights. Through the integration of embedded systems, software analytics, and sustainability-focused visualisation, the project aims to make carbon savings more visible, measurable, and understandable to users.

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